

1 Sharing sunlight between trackers and crops: dynamic
2 light-sharing control for agrivoltaics under a
3 food-security floor on the world's staple croplands

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6 **Abstract**

Single-axis trackers let agrivoltaic arrays shift the daily rotation from maximum electricity towards letting light through (light sharing). We quantify its value under a food-security floor of 90% of open-field yield. A differentiable simulator couples solar geometry, trackers, PV output (PVGIS benchmark: $r = 0.97$), crop light, microclimate, soil water and the SIMPLE crop model. The crop response to shade is its least certain part, so we learned it from data: a hierarchical model fitted to 428 observations from 58 field studies predicted held-out studies with a 28% lower error than proportional loss (RMSE 0.31 against 0.43), and gradient descent through the simulator turned it into an ensemble of plausible responses. In a deployable protocol (density chosen from each site's history, held-out sites and years, 400 crop-site pairs, baseline to +3 °C), light-sharing rules allowed arrays three times as dense as static arrays at equal retention and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24; a phenology rule needing no forecast was the best. A neural controller trained through the simulator added only 0.01–0.05 LER, and neither training on the ensemble nor a farm-measurable biomass signal changed this. The decision that mattered was the design: a density chosen under the mean response met the floor at 62–63% of sites when another plausible response held (oracle 73–79%), and a density meeting the floor in 75% of plausible responses raised this to 71–73% for 0.04 LER. The simulator is too mild for maize and soybean, so gains for these crops are probably overestimated. Equipping 5% of staple cropland would give about 14 PWh yr⁻¹ but needs about 10 TWp of modules.

7 *Keywords:* Agrivoltaics, Light sharing, Solar tracking, Food-energy nexus,
8 Crop modelling, Differentiable simulation, Hierarchical Bayesian model

9 1. Introduction

10 Two of the defining challenges of this century compete for the same re-
11 source: land. The energy transition requires terawatts of new solar pho-
12 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
13 to two hectares per megawatt. Food production must rise while crops are
14 exposed to more frequent heat waves and droughts. Warming alone is pro-
15 jected to reduce global yields of maize, wheat, rice and soybean by several
16 per cent per degree (Zhao et al., 2017), and a new generation of climate and
17 crop models shows these impacts emerging earlier than previously expected
18 (Jägermeyr et al., 2021). Croplands are also where solar power potential is
19 greatest (Adeh et al., 2019). They are flat, sunny, accessible and close to
20 rural electricity demand, so converting farmland into solar farms is tempting
21 and controversial at the same time.

22 Agrivoltaic (AV) systems combine crops and PV modules on the same
23 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
24 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
25 a hectare can deliver more food plus energy than the two uses side by side,
26 with land-equivalent ratios (LER) well above one in field trials and mod-
27 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
28 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
29 climate. It lowers canopy temperature and evaporative demand, conserves
30 soil moisture and improves water-use efficiency, especially in drylands (Mar-
31 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
32 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a candi-
33 date adaptation measure against heat and drought stress, in addition to a
34 mitigation measure through renewable electricity. The same shade, however,
35 also reduces the light available for photosynthesis. Regulators increasingly
36 require AV systems to retain most of the reference crop yield. The German
37 pre-standard DIN SPEC 91434, for example, requires at least 66% of the
38 reference yield (DIN, 2021), and other schemes and many farmers set stricter
39 floors. How much electricity can be produced without sacrificing food secu-
40 rity depends on the crop, the local climate, the year and the growth stage.

41 Most AV designs fix the geometry of the array, i.e. the ground coverage
42 ratio (GCR, module area per ground area) and the tilt or tracking rule, from
43 generic rules of thumb. Single-axis trackers add a degree of freedom: the

modules can be rotated away from the sun to let light through when the crop needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the sun to maximise electricity. Choosing the rotation every day is a sequential decision problem. The value of light depends on the crop’s phenological stage, soil water and upcoming weather, and the effect of a decision appears in the harvest months later. Recent work in this journal has used evolutionary multi-objective optimisation to lay out PV arrays on farmland (Zhang et al., 2026). Machine learning for crop management has mainly used reinforcement learning with black-box simulators (Gautron et al., 2022), which is sample-inefficient and hard to scale to global, multi-decadal assessments.

Two questions follow. First, how much of the electricity that an agrivoltaic system can produce under a food-security floor comes from the array density, and how much from dynamic light-sharing control of the trackers? Second, how robust is that answer to the weakest part of any such study, the crop response to shade, and does a controller learned from a simulator add anything to a well-designed hand-written rule? Answering them requires a simulator that couples the physics of light, PV, microclimate and crop growth, a protocol in which all controllers face the same design constraints, a check of the shade response against field data, and enough sites and seasons to separate real effects from noise.

This paper makes three contributions. The first is a quantification of light-sharing control under a yield floor. Light sharing is parameterised by one daily level between full backtracking and the rotation that minimises the shadow on the crop, and we compare static arrays, a seasonal schedule, a rule that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a deployable evaluation protocol. The array density of every site is chosen from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three training seeds and three spatial splits. The third is a differentiable agrivoltaic simulator that supports both, coupling hourly solar geometry, single-axis trackers with backtracking, row-to-row shading and PV electricity, the direct and diffuse light reaching the crop, canopy temperature and reference evapotranspiration under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty. All components are differentiable tensor operations (Karniadakis et al., 2021). The PV part is benchmarked against PVGIS. The crop response to shade is learned from the open data of a meta-analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato

and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023), and propagated through the simulator as an ensemble.

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever. At the same mean yield retention of 90%, arrays with light-sharing control carry about three times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section 3.3).
3. The crop response to shade can be learned from data. A hierarchical model fitted to 428 observations from 58 field studies predicts held-out studies with an error 28% below that of proportional loss, with crop type as the transferable information. The simulator is close to these curves for cereals and potato but too mild for maize and soybean, so the gains for those two crops are probably overestimated (Section 3.7).
4. A learned controller adds only a small gain over the best rule, 0.01–0.05 LER. Training it on an ensemble of shade responses, or giving it a farm-measurable biomass signal, does not change this (Section 3.8).
5. The bottleneck is the uncertainty of the crop response that enters the design. A density chosen so that the floor holds in 75% of the plausible responses raises the share of compliant sites from 62–63% (design under the mean response) to 71–73% (oracle: 73–79%) for 0.04 LER (Section 3.8).
6. At the yield floor, agrivoltaics on 5% of the world’s staple cropland could supply about 14 PWh yr^{−1}, but only with about 10 TWp of modules, several times the installed PV capacity of the world (Section 3.10).

2. Materials and methods

2.1. Global cropland sites, weather and crop calendars

We studied five staple crops that together cover about 670 Mha of harvested area and 580 Mha of physical cropland: wheat, rice, maize, soybean

118 and potato. Harvested areas (rainfed and irrigated) were taken from the
 119 SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1°
 120 cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn
 121 without replacement with probability proportional to harvested area. An un-
 122 weighted average over the sample therefore approximates an area-weighted
 123 average over global production. The 400 crop–site pairs lie in 381 distinct
 124 cells in 50 countries (Fig. 2). Sowing and maturity dates came from the
 125 GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat
 126 we used the winter-wheat calendar unless the coldest-month mean temper-
 127 ature was below -8°C , in which case the spring-wheat calendar was used.
 128 Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks,
 129 2018): global and diffuse shortwave irradiance, maximum, minimum and
 130 dew-point temperature, precipitation and wind speed at 2 m. Each site con-
 131 tributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days
 132 after sowing, which gives 7600 site-seasons.

133 Climate change was represented by the global warming levels $+1.5$, $+2$
 134 and $+3^\circ\text{C}$ relative to 2001–2020. Daily maximum, minimum and dew-point
 135 temperatures were shifted uniformly, which keeps relative humidity approx-
 136 imately constant and raises the vapour-pressure deficit. The atmospheric
 137 CO_2 concentration was set to 390, 450, 500 and 600 ppm for the baseline and
 138 the three warming levels. Precipitation and irradiance were left unchanged.
 139 This delta approach isolates the thermal signal of warming; its limitations
 140 are discussed in Section 4.5.

141 2.2. A differentiable agrivoltaic simulator

142 The simulator advances all site-seasons in parallel with a daily time step
 143 for the crop and an hourly time step for radiation and PV (Fig. 1). All equa-
 144 tions are implemented as PyTorch tensor operations (Paszke et al., 2019), so
 145 that yields and electricity are differentiable with respect to the design and
 146 control variables.

147 *Hourly irradiance.* Daily global and diffuse irradiance were disaggregated
 148 into hourly values with the Collares-Pereira and Rabl (1979) and Liu and
 149 Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance
 150 on the horizontal is the difference between the two. Solar position follows
 151 the standard declination and hour-angle equations.

152 *Array geometry and PV electricity.* We consider elevated horizontal single-
 153 axis trackers with north–south axes. The axis height is 1.25 module widths,

154 e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The ro-
 155 tation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width
 156 divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this
 157 keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the
 158 rows when the modules are horizontal, which is the stow position during field
 159 operations. Let ψ be the solar zenith angle projected onto the east–west ver-
 160 tical plane. The energy-optimal rotation with backtracking (Lorenzo et al.,
 161 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ oth-
 162 erwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on
 163 to the sun, which minimises the shadow on the crop. Given a light-sharing
 164 level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

165 Plane-of-array irradiance combines three terms: beam irradiance on the mod-
 166 ule, reduced by the fraction of the module shaded by the neighbouring row,
 167 $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected
 168 light (albedo 0.2). The cell temperature follows the NOCT model (45°C),
 169 with an hourly air temperature interpolated sinusoidally between the daily
 170 extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with
 171 $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature
 172 losses and system losses. The installed capacity is $2.1 \text{ g MWp ha}^{-1}$. A con-
 173 ventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the
 174 energy reference E_{PV} .

175 *Validation of the PV model.* We compared the simulated specific yield of an
 176 isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2
 177 (European Commission, Joint Research Centre, 2022), for a horizontal north–
 178 south single-axis tracker with 14% losses, at 30 randomly chosen sites (29
 179 returned data). Across sites, the simulated and PVGIS specific yields were
 180 correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean
 181 absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit and
 182 with the different irradiance databases (NASA POWER vs. SARA2, ERA5
 183 and NSRDB). Electricity estimates are therefore, if anything, conservative.

184 *Light and microclimate at the crop.* Averaged over the row pitch, the fraction
 185 of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The
 186 fraction of sky diffuse reaching the ground is a two-dimensional isotropic
 187 sky-view factor. We pre-computed it for 25 values of g and 13 rotation

188 angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly
 189 (differentiably). The daily ratio s of the irradiance at the crop to that of the
 190 open field defines the crop’s shading. The canopy maximum temperature is
 191 lowered by $\kappa(1 - s)$.

192 κ is uncertain. Field studies under PV modules report small reductions
 193 of daytime crop or canopy temperature and little change in air temperature
 194 (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds
 195 calibrated on air temperature do not transfer directly to canopy temperature
 196 (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate
 197 $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$
 198 for all methods (Section 3.2). Nocturnal warming under the modules is not
 199 represented. The reference evapotranspiration under the modules, ET_0 , is re-
 200 computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998)
 201 from the reduced radiation and canopy temperature. Soil water conserva-
 202 tion under the modules (Hassanpour Adeh et al., 2018; Elamri et al., 2018)
 203 therefore emerges from the model, but rain redistribution by the modules
 204 (Marrou et al., 2013a) is ignored.

205 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)
 206 with the species parameters of its Table 1a (Table 1). Daily biomass growth
 207 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

208 where $R_{\text{crop}} = sR$ is the radiation received by the crop. The terms are as
 209 follows.

- 210 • f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends
 211 only on thermal time, with senescence accelerated by heat and drought
 212 stress. Shade therefore reduces the daily biomass increment but not
 213 leaf-area development.
- 214 • f_{Heat} decreases linearly between the heat threshold T_{heat} and the ex-
 215 treme temperature T_{ext} of the canopy maximum temperature.
- 216 • $f_{\text{Water}} = 1 - S_{\text{water}} \text{ ARID}$, where the agricultural reference index for
 217 drought (Woli et al., 2012) is computed from a daily soil-water balance
 218 with curve-number runoff, drainage and root water uptake. Irrigated
 219 fields have $f_{\text{Water}} = 1$.

220 SIMPLE has a fixed harvest index and therefore ignores the high sen-
 221 sitivity of grain number to shading around flowering (Fischer, 1985; An-
 222 drade et al., 1999). It also ignores the differences in shade tolerance be-
 223 tween crop types. In the meta-analysis of Laub et al. (2022), maize and
 224 grain legumes lost yield strongly even at low shade levels, whereas C3 ce-
 225 reals and tuber crops initially lost less than proportionally; shading trials
 226 also show marked effects of shade on maize development (Ramos-Fuentes
 227 et al., 2023). We added a stage-dependent penalty. During a critical window
 228 ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$
 229 is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h
 230 is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al.
 231 (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice
 232 0.5, for which no specific curve was available) and lowest for potato (0.2).
 233 These values are assumptions that reproduce this ranking; they are not cal-
 234 ibrated estimates. Section 3.6 shows the resulting shade-response curves.
 235 Because RUE is linear, SIMPLE does not credit the higher light-use effi-
 236 ciency of shaded canopies under more diffuse light. The penalty, conversely,
 237 makes shading around flowering more damaging than proportional. The net
 238 effect is closer to the field evidence than either assumption alone.

239 The thermal-time requirement to maturity was calibrated for every site as
 240 the mean thermal time between the GGCM sowing and maturity dates under
 241 the 2001–2020 climate, and the canopy parameters were scaled accordingly.
 242 The cultivar is thus adapted to the present climate, and warming shortens the
 243 season. Growth stops smoothly once the thermal-time requirement is met.
 244 Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site
 245 is represented by a rainfed and an irrigated sub-field in proportion to its
 246 SPAM irrigated fraction. Each sub-field carries its own array, so both yield
 247 and electricity are area-weighted over the two sub-fields.

248 2.3. SALS: simulation-trained adaptive light sharing

249 SALS learns a *control network* $u_t = \pi_{\omega}(\mathbf{x}_t; g)$ (Algorithm 1). It is a
 250 multilayer perceptron with two hidden layers of 64 units and a bounded-logit
 251 output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero
 252 at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs
 253 are the information available on day t :

- 254 • the crop and soil state: relative thermal time TT/T_{sum} , radiation in-
 255 terception f_{Solar} , relative soil water and the previous day’s ARID;

- a *forecast* of the day’s maximum temperature (relative to T_{heat} and T_{opt}), irradiance and ET_0 ;
- static information: the irrigation status of the sub-field, the array’s GCR g , latitude, day of season and the crop.

Forecasts are perturbed with random errors, independently every day: Gaussian with a standard deviation of 2°C for maximum temperature, and multiplicative with standard deviations of 20% for irradiance and 15% for ET_0 . The physics always uses the true weather. The same forecast errors are applied to the rule-based controllers. The controller is conditioned on g . During training, each season is simulated with a GCR close to the density that is sustainable at its training site. This is the largest value on a coarse grid $(0.05, 0.10, \dots, 0.50)$ at which seasonal sharing keeps the mean retention of the site’s training seasons at or above ρ , plus a uniform perturbation in $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to control the densities that design selection will later pick. Outside the cropping season the trackers backtrack for maximum electricity.

For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

which combines an exact (hinge) and a quadratic penalty for the food-security floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the exact gradient of Eq. (3) is obtained by back-propagation through the 365 daily steps of every season (back-propagation through time). This is the key difference from reinforcement learning with black-box simulators (Gautron et al., 2022), which estimates gradients from sampled returns.

Each minibatch contains 512 site-seasons, each simulated as a rainfed and an irrigated sub-field, drawn from one of the four climates in turn. Seasons with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training and evaluation. The network is fine-tuned through the simulator with Adam (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in a numerically stable form, so that gradients remain finite even under extreme heat.

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
- 2: **for** step = $1, \dots, N$ **do**
- 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
- 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5: **for** day $t = 1, \dots, 365$ **do**
- 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
- 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
- 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
- 10: back-propagate through the season; Adam update of ω

289 *Initialisation.* Trained from random weights, the controller converged to
290 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
291 the seasonal-sharing rule, because the gradient of electricity with respect to u
292 is small and of the same sign for most days of a season. We therefore initialise
293 the network by behaviour cloning of the phenology rule of Section 2.4: binary
294 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
295 1.7 million day-states visited by the rule in the training seasons, with 1500
296 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
297 policy away from this initialisation wherever doing so improves the objective.
298 Consequently, the gain of SALS over the phenology rule measures what fine-
299 tuning through the simulator adds to that rule, not what can be learned
300 without it.

301 *2.4. Deployable design selection and baselines*

302 A real array has one fixed density, which must be chosen before it is built.
303 Every method, SALS included, therefore uses the same *history-based design*
304 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
305 for which the method’s mean yield retention over the site’s 2001–2014 sea-
306 sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
307 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in

at least 80% of the historical seasons. The historical seasons of the test sites are never used for training, and the weather of the evaluation years is never used for design. We compare four controllers:

- *AV static*: backtracking trackers without light sharing ($u = 0$);
- *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the cropping season;
- *Stress rule*: an expert rule that shares light ($u = 1$) during the main growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules track the sun and shade the crop;
- *SALS*: the learned controller.

The open field and a conventional PV plant without crops ($\text{LER} = 1$ each) are the references. Because the design rule is identical for all controllers, differences between methods reflect the quality of control and the array density that each controller can sustain under the food-security floor. We also compare the controllers at identical GCR over the whole grid (matched-density frontiers).

2.5. A shade-response model learned from field data

The simulator needs a crop response to shade that is supported by observations. We learned one from the open data of the meta-analysis of Laub et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observations of yield relative to an unshaded control at relative shade reductions of 0–90% from 58 studies, with crop and crop type, experiment type (artificial shading or intercropping), climate zone, irradiation and shade type. Relative yield is modelled as $y(s) = \exp(-\beta s^\gamma)$ with Student- t errors ($\nu = 4$). We compared six models: proportional loss $y = 1 - s$; one curve for all crops; one curve per crop type without pooling; one curve per crop type with partial pooling (Gaussian shrinkage of $\log \beta$ and $\log \gamma$ towards a common value, scale 0.5); partial pooling with covariates on $\log \beta$ (standardised irradiation, subtropical zone, solar-panel shade and intercropping); and gradient-boosted trees on the same inputs. Fits use L-BFGS. Models were compared by leave-one-study-out cross-validation, in which every study is held out once, which tests the prediction of a new study and not of a new observation in a known study. Intervals for the crop-type curves and covariate effects come from a cluster bootstrap over studies (150 resamples).

343 2.6. Shade-response uncertainty: gradient calibration, robust controllers and 344 design

345 The crop’s response to shade is the least certain part of the simulator,
346 so we treat it as an unknown with a distribution. Two parameters per crop
347 enter the simulator: a radiation-use compensation c (growth is multiplied by
348 $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop)
349 and a multiplier h on the flowering-window harvest-index penalty. Because
350 the simulator is differentiable, both can be fitted by gradient descent through
351 whole simulated seasons to published yield–shading observations. The ob-
352 servations are the relative yields at 20% and 40% shading of the data-driven
353 curves of Section 2.5 for the five crops (median, with an error equal to the
354 width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice
355 takes the curve of C3 cereals, which contains no rice), the field trials on
356 rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021;
357 Ramos-Fuentes et al., 2023) (each season of each trial as a separate obser-
358 vation, error 0.06–0.08) and the unaffected harvest index of shaded maize
359 (Ramos-Fuentes et al., 2023). We call the resulting ensemble E2. An earlier
360 ensemble, E1, was fitted in the same way to curves read by hand from the
361 figures of Laub et al. (2022) and was used to train the robust and adaptive
362 controllers below. Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and $h \sim \mathcal{N}(1, 0.5^2)$, truncated
363 at zero. We fitted 12 ensemble members by randomised maximum a pos-
364 teriori estimation: every member sees the observations perturbed by their
365 errors and a prior draw perturbed by its width, and is optimised with Adam
366 (40 steps, 40 training-block seasons per crop and member, shading of the
367 static arrays cached). The ensemble approximates the posterior of the shade
368 response and is used in three ways.

369 *Robust and adaptive controllers.* SALS-robust has the architecture and
370 training of SALS (Section 2.3), but every training season is simulated with the
371 response of a randomly drawn ensemble member that the controller does not
372 observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible
373 densities differ between members. SALS-adaptive additionally observes the
374 realised biomass of the crop under the modules relative to an open reference
375 plot, a quantity that can be measured on the farm by remote sensing or by
376 a reference plot. A feedback rule that shares light whenever this ratio falls
377 below a target serves as the hand-written counterpart; its target (1.0) was
378 chosen on training-block seasons under the ensemble.

379 *Uncertainty-aware design.* The design rule of Section 2.4 chooses the
380 largest GCR whose historical retention meets the floor. Under shade-response

uncertainty we choose the largest GCR that meets the floor in at least 75% of five ensemble members, and compare it with a design under the ensemble mean and with an oracle design under the true response. Evaluation of this rule is leave-one-out: the truth is the held-out member and the design uses the other four.

Stress tests. Controllers and rules are also evaluated under four fixed responses, the proportional response, the response calibrated to the meta-analysis, the response fitted to the field trials and a harsh response (proportional growth and a 1.5 times larger harvest-index penalty). The proportional and harsh responses lie outside the range of the ensemble.

2.7. Evaluation protocol, statistics and upscaling

Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to training (70%) or test (30%), so that no test site has a training site in the same block. Training uses the 2001–2014 seasons of the training sites. Design selection uses the 2001–2014 seasons of the test sites. Evaluation uses the 2015–2019 seasons of the test sites. The same site-seasons are analysed in all four climates.

We report the following metrics:

- the mean electricity ($\text{MWh ha}^{-1} \text{yr}^{-1}$) and the relative electricity e ;
- the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- season-level compliance (share of seasons with $r \geq \rho$) and site-level compliance (share of sites whose mean test-season retention is $\geq \rho$);
- the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011).

Statistical comparisons use site means as the unit of analysis. We report paired Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence intervals of the mean difference. SALS was trained with three random seeds on the main split, and with one seed on each of two further spatial splits.

For the global estimate, every site is evaluated in the split in which it is held out. The mean electricity and yield per hectare of each crop are scaled to the *physical* area of that crop in SPAM 2010, which avoids double counting multiple-cropped land (Waha et al., 2020). We report the installed capacity (TWp), the capacity factor and a supply curve.

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in $^{\circ}\text{C}$, thermal times in $^{\circ}\text{C d}$, RUE in gMJ^{-1} ; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

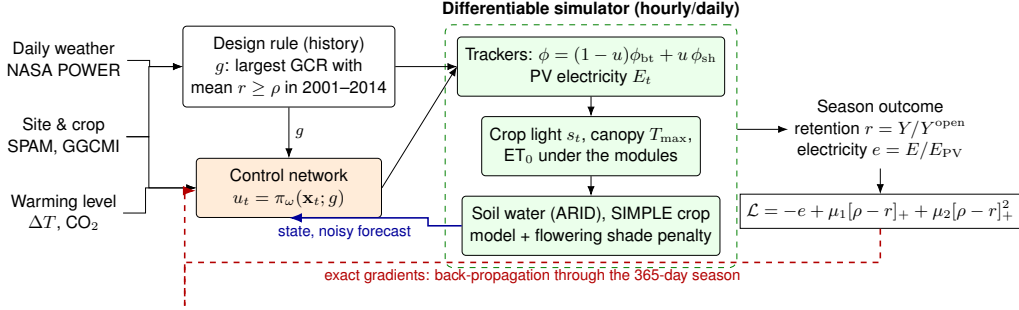


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

3. Results

3.1. Electricity and food on held-out sites and years

Table 2 compares the four controllers on the 92 held-out sites of the main split (seasons sown in 2015–2019), after each site’s density has been chosen from its 2001–2014 seasons with the same rule for all controllers. By construction all methods keep the mean yield retention close to 90% (0.904–0.917), so differences in electricity are differences in the density that each controller can sustain under this floor.

Static arrays are the weakest. Without light sharing, the floor limits the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$, 16% of a conventional PV plant per hectare (LER 1.07). Rotating the

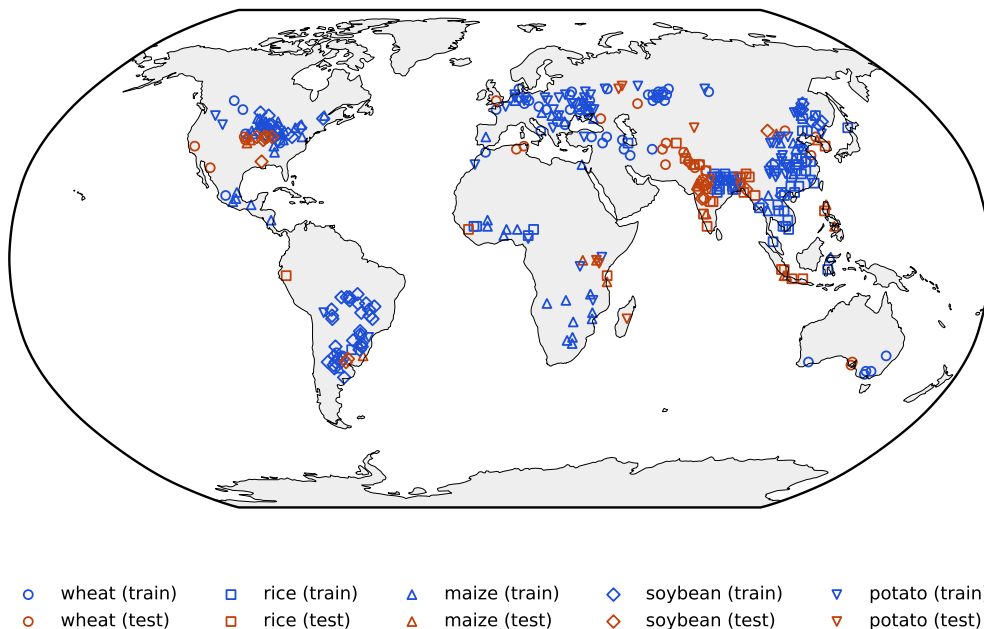


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

modules to let light through during the cropping season (seasonal sharing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule that shares light only while the crop is growing (phenology rule) reaches $465 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.24). An expert rule that additionally shades the crop on forecast heat or drought days (stress rule) is worse than the phenology rule (LER 1.18): its density is limited to 13% because it reduces light at times when the floor is most binding.

The learned controller is best by a modest, consistent margin. SALS sustains a density of 17% (similar to the phenology rule), produces $510 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ and reaches an LER of 1.270 at the same retention. Compared with the phenology rule this is +0.038 in LER (95% CI over sites 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Table A.11). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows the same picture at matched densities: the learned policy lies on or slightly

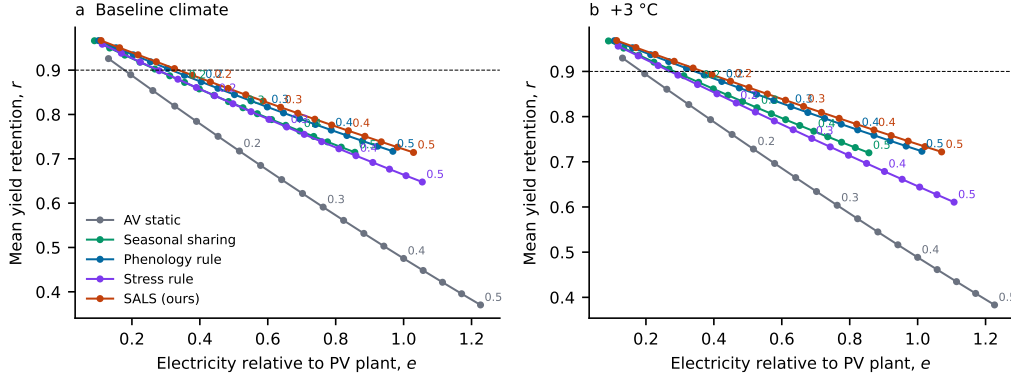


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

above the phenology rule over the whole density range, both lie well above seasonal sharing and the stress rule, and static arrays are far below.

What the floor does and does not guarantee. Enforcing the floor on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained design) raises the season-level compliance to 90–94% for the light-sharing controllers. It costs about 0.065 in LER for SALS (1.205), which remains the best controller (phenology rule 1.192; Section 3.2).

3.2. Robustness

Seeds and splits. Three training seeds of SALS give LERs of 1.270, 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at +3 °C. On two further spatial splits, in which different blocks of sites are held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under the baseline climate and by 0.052 and 0.039 at +3 °C (main split: 0.034 and 0.032). In every split SALS ranks first, the phenology rule second and static arrays last.

Canopy cooling. The canopy-cooling coefficient κ is the most uncertain microclimate parameter. We repeated the whole protocol, includ-

ing the historical design step, with $\kappa = 0, 1$ and 2°C . Electricity changes with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and 549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at 0.905 by the design rule and the ranking is unchanged. SALS exceeds the phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate (0.022, 0.032 and 0.051 at $+3^\circ\text{C}$).

Which ingredients matter. Two ablations test the design of SALS at the same training budget. Masking the stress inputs (soil water, ARID, and the maximum temperature, irradiance and ET₀ forecasts) leaves the LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297 at $+3^\circ\text{C}$). Training only on the baseline climate gives 1.268 and 1.291. The improvement over the rules therefore does not depend on stress-aware inputs or on training in warmed climates.

3.3. Floor conditioning, chance-constrained training and forecast errors

One network for all floors. Regulations and farmers set different floors. The network of the main experiments was trained at $\rho = 0.9$ only. We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical design step for each floor (Table 3). Its LER exceeds that of the phenology rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037, 0.032 and 0.033 at $+3^\circ\text{C}$), so the benefit carries over to other floors without retraining. A network that received the floor as an additional input and was trained over all four floors did not improve on it (differences in LER ≤ 0.001). Because the design step already adapts the density to the floor, conditioning the controller on the floor is not needed.

Chance-constrained training. We also trained a network whose objective penalises a batch share of seasons below the floor that exceeds 20%. It did not improve the tail. With the mean-based design step its LER was 1.263 (baseline) and 1.290 ($+3^\circ\text{C}$) against 1.270 and 1.297, and with chance-constrained design it reached season-level compliance of 92.6% against 92.0% and an LER of 1.198 against 1.205. The season-level guarantee is obtained by the design step (Section 3.1), not by the training objective.

Forecast errors. We raised the forecast errors seen by the controllers to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and of the phenology rule was identical at all four error levels to three decimals (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and 1.416). Neither controller uses the weather forecast: both rely on the crop’s thermal time, which is observable from on-farm temperature records. The

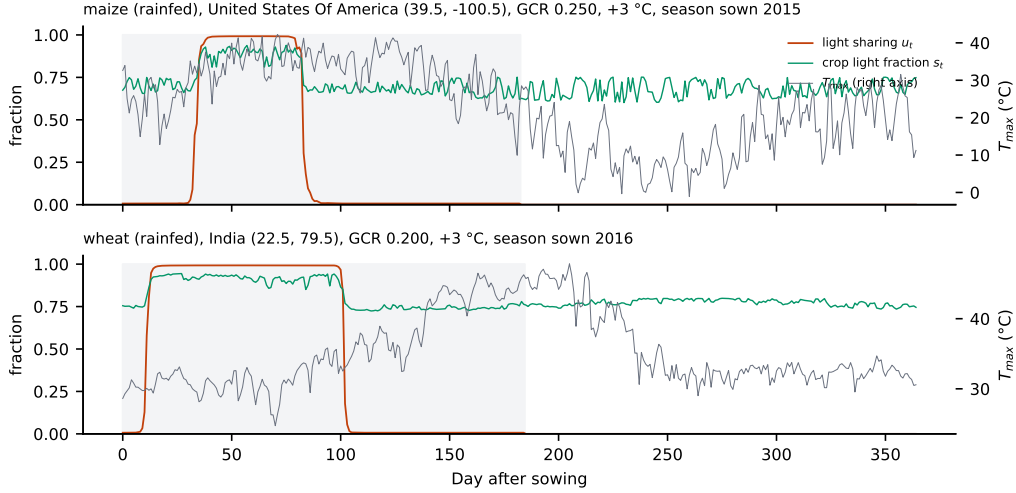


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

500 stress rule, which depends on the forecast of maximum temperature, degrades
 501 with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and
 502 its LER from 1.432 to 1.418 as the errors triple.

503 3.4. What the learned policy does

504 Averaged over the rainfed test seasons of the main split at a density of 0.3,
 505 the light-sharing level of SALS is a smooth function of relative thermal time.
 506 It is close to zero in the first 10% of the thermal time required for maturity,
 507 rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and 70%, and
 508 falls through 0.58 at 80–90% to about zero at maturity. The plateau covers
 509 the window in which our crop model penalises shading through the harvest
 510 index. The ramps at both ends let the modules produce electricity when
 511 additional light is worth least to the crop. Fig. 4 shows two example seasons.
 512 The phenology rule, which is also the initialisation of the network, shares
 513 light during the whole growing period. Fine-tuning through the simulator
 514 found that sharing can be reduced at the edges of the season: SALS gains its
 515 advantage by yielding light when it matters and collecting electricity when
 516 it matters less.

517 The policy also shares less light on days with a forecast heat stress: the
 518 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 (+3 °C)
 519 on other days, and 58–63% of sites show a lower level on heat days. This
 520 cannot be read as a stress response, however. Heat days occur late in the
 521 season, when the phenological ramp is already falling, and a policy without
 522 any stress input shows a similar contrast (0.37 vs 0.57 in a separate check).
 523 The ablation above confirms that stress information does not improve the
 524 result.

525 3.5. Crops and regions

526 The benefit of light sharing differs between crops (baseline climate, SALS).
 527 Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and
 528 577 MWh ha⁻¹ yr⁻¹; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31),
 529 and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20
 530 for both). The stress rule has a lower LER for every crop (1.26 for soybean,
 531 1.14–1.21 for the others).

532 The floor is not met evenly. Production-weighted retention is 0.888 for
 533 soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose
 534 more than low-yielding ones. Regionally, Latin America and Oceania (4 sites)
 535 combine the highest density (GCR 0.41) and electricity (1090 MWh ha⁻¹ yr⁻¹)
 536 with a production-weighted retention of only 0.862. Africa and the Middle
 537 East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention
 538 of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902
 539 (67% of sites) at a density of 0.14. A regulatory floor enforced per site would
 540 therefore need a region- and crop-specific design step, not a global one.

541 3.6. Model behaviour and validation

542 **PV output.** The simulated specific yield of an isolated tracker row agrees
 543 with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29
 544 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation
 545 of 5.5% (Section 2.2).

546 **Shade response.** Fig. 5 shows the simulated relative yield of rainfed
 547 crops under static arrays as a function of the season-mean shading of the crop.
 548 Yield declines close to proportionally with shading: 84–89% relative yield at
 549 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat
 550 loses most and potato and maize least. This response is steeper than the *less*
 551 *than proportional* losses that Laub et al. (2022) report for small shade levels
 552 of C3 cereals and tubers, and it does not reproduce their finding that maize

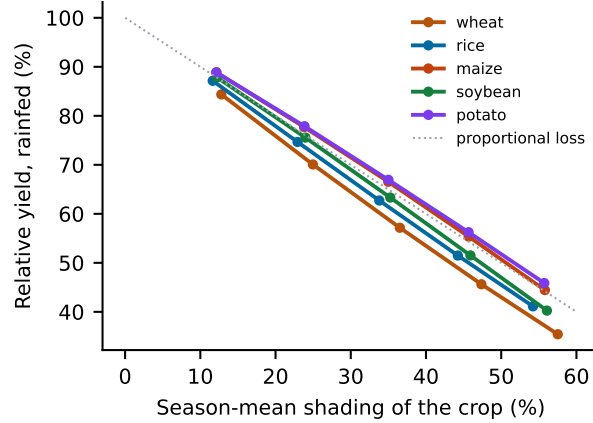


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

and grain legumes are more shade sensitive than C3 cereals. In the simulator, shading also relieves heat and drought in rainfed maize and soybean, which offsets the higher harvest-index penalty we assigned to them. The absolute effect of shading on each crop is therefore uncertain (Section 4.5).

Comparison with field trials. We fitted one radiation-use compensation coefficient per crop to the yield–shading curves of Laub et al. (2022) using training-block sites only (Methods, calibration script in the repository). The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the proportional model already lies within about 0.02 of the C3-cereal and tuber curves, and it cannot lower the maize and legume responses below the proportional case (root-mean-square error 0.16 for maize and 0.08 for soybean). Against independent field trials (Table 4) the model underestimates retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al. (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was not affected by shade. The model is therefore conservative for food retention with respect to these trials, except for the stronger maize losses of the meta-analysis. Table 5 shows the consequence for the design results: with the calibrated response the allowed density and the LER of every method rise (phenology rule 1.28 against 1.23), and with a response fitted to the

field trials the LER of the phenology rule is 1.62. The phenology rule is the best method in all three cases, and light sharing adds LER +0.17, +0.19 and +0.30 over static arrays (conservative, calibrated, field-fitted); only the order of the seasonal and stress rules changes. The absolute gains reported in this paper are thus lower bounds with respect to these trials and not upper bounds.

3.7. A shade-response model learned from field data

The curve learned for each crop type predicts new studies better than a proportional loss (Table 6). In leave-one-study-out cross-validation the root-mean-square error falls from 0.43 for $y = 1 - s$ to 0.37 for one curve for all crops and to 0.31 for crop-type curves, a reduction of 28% relative to the proportional loss. Partial pooling is as good as separate curves (0.312 against 0.313), and the covariates add almost nothing (0.311). Gradient-boosted trees on the same inputs are worse (0.321) than the parametric hierarchical model, as expected for 58 studies. Crop type is therefore the information that transfers between studies. None of the covariate effects on $\log \beta$ is clearly different from zero (irradiation -0.10 , 90% interval -0.32 to 0.09 ; subtropical zone -0.37 , -0.73 to 0.07 ; intercropping 0.16 , -0.48 to 0.45). The estimated effect of solar-panel shade is not interpretable, because all 46 panel observations are of leafy vegetables, mainly lettuce, and the effect is confounded with crop type.

The learned curves show a clear ordering of crop types (Table 7). At 20% shading C3 cereals and tubers lose less than proportionally (relative yield 0.93), whereas maize (0.73) and grain legumes (0.71) lose more; at 40% shading the values are 0.72 and 0.71 against 0.42 and 0.40. Intervals are wide for maize (0.27–0.59 at 40%) and for tubers, which rest on two studies. The simulator’s responses are close to the data for cereals and potato but, under both calibrated ensembles, too mild for maize and soybean at 40% shading (0.75 and 0.48 for E2 against 0.42 and 0.40 in the data). The calibration cannot lower the maize response further, because the simulated shading also relieves heat and drought stress and the compensation and harvest-index parameters are bounded below at zero. The field trials of maize (Ramos-Fuentes et al., 2023) are also milder than the meta-analysis. For maize and soybean the gains reported in this paper are therefore probably overestimated.

608 3.8. Learning under shade-response uncertainty

609 **Calibration.** Gradient descent through the simulator fitted the five crops
 610 to the data-driven curves and the field trials (ensemble E2, Table 8). The
 611 data identify two features: the harvest index of maize is hardly affected by
 612 shade (multiplier 0.16 ± 0.13 , $c \approx 0.14$), as observed by Ramos-Fuentes et al.
 613 (2023), and soybean is more shade sensitive than the cereals (compensation
 614 0.05, harvest-index multiplier 1.66), as in Laub et al. (2022). Wheat has the
 615 largest compensation (0.91 ± 0.31) and the largest uncertainty.

616 **Learned controllers.** Controllers were trained on E1 and evaluated
 617 under four fixed responses. Under every response the learned controller stays
 618 better than the best hand-written rule, but only slightly (Table 9): the LER
 619 gain of SALS over the phenology rule is +0.014, +0.015, +0.033 and +0.010
 620 under the proportional, calibrated, field-fitted and harsh responses. Training
 621 on the ensemble did not help: SALS-robust differs from SALS, which was
 622 trained on the proportional response only, by at most 0.005 in LER, and
 623 SALS-adaptive, which sees the measured biomass ratio, by at most 0.008.
 624 The feedback rule is worse than the phenology rule by 0.025–0.064 in LER.
 625 A plausible reason is that the biomass ratio does not reveal the harvest-index
 626 loss from shading around flowering, so the information that the controller
 627 could use to adapt is not in the measurement.

628 **Design under uncertainty.** The larger risk is in the design step (Ta-
 629 ble 10). When the density is chosen under the mean response of E2 and
 630 the true response is a different plausible one, the share of sites whose mean
 631 retention reaches the floor is 63% (phenology rule) and 62% (SALS), against
 632 79% and 73% for an oracle design. Choosing the density so that the floor is
 633 met in at least 75% of the plausible responses raises site compliance to 73%
 634 and 71%, which recovers most of the gap to the oracle, at a cost of 0.042
 635 and 0.040 in LER relative to the design under the mean response. With the
 636 earlier ensemble E1 the same design restored the oracle level (76% and 74%,
 637 against 58% and 55% under the mean) at a cost of 0.029 and 0.023 LER. For
 638 the proportional and harsh responses, which lie outside the ensembles, the
 639 design does not restore compliance (about 20–24% of sites under E2), so the
 640 guarantee holds only within the range of responses that the data support.

641 3.9. Agrivoltaics under warming

642 Under warming, the open-field production of the held-out sites (Fig. 6)
 643 falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soy-
 644 bean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production

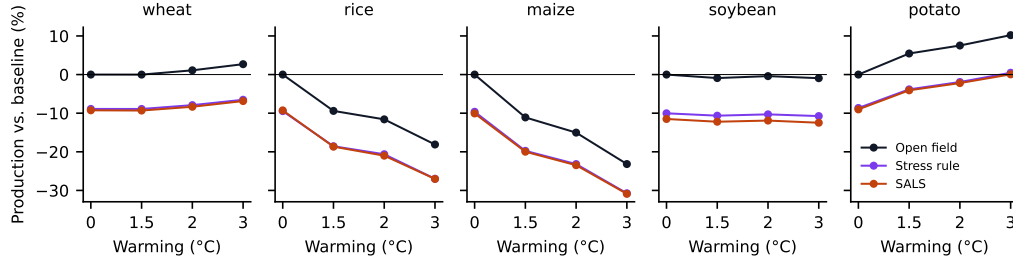


Figure 6: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

with light-sharing control follows the open field at about 90% of its level, so it inherits the same climate impacts (maize -31% , rice -27% relative to the baseline open field at $+3^\circ\text{C}$). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265), because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 $\text{MWh ha}^{-1} \text{yr}^{-1}$). The margin of SALS over the rules does not grow with warming (Table A.11).

3.10. Global potential and what it would take

Every site is evaluated when it is held out (219 of the 400 crop-site pairs, in at least one split), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr^{-1} under the baseline climate and 16.0 PWh yr^{-1} at $+3^\circ\text{C}$, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr^{-1} , and static arrays 6.4 and 6.9 PWh yr^{-1} . For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at $+3^\circ\text{C}$), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.7 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 7): the best 2% of the staple cropland would supply 16 PWh yr^{-1} and the best 5% 40 PWh yr^{-1} with

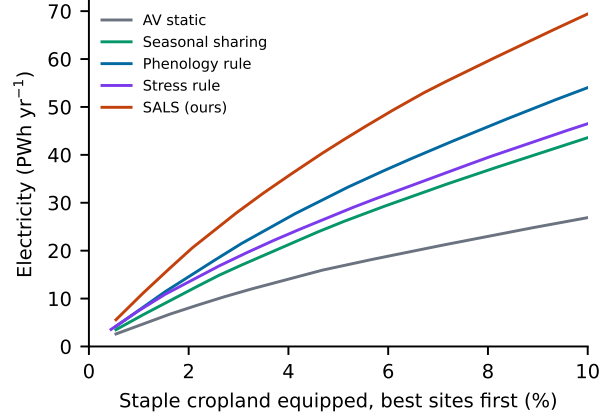


Figure 7: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

671 SALS. The best tenth of the sites are mostly rainfed maize and soybean sites
672 in the Americas and in southern and eastern Africa (7% mean irrigated frac-
673 tion against 29% overall), which are also among the regions where the floor
674 is hardest to meet (Section 3.5). Siting by electricity alone would therefore
675 concentrate agrivoltaics where food-security risk is highest.

676 4. Discussion

677 4.1. What the results establish

678 Four findings stand on consistent evidence. First, sharing light with the
679 crop is the main lever on the electricity that tracker-based agrivoltaic arrays
680 can produce under a yield floor. With static trackers the floor limits arrays
681 to about 6% ground coverage and an LER of 1.07. Any light-sharing control
682 allows about three times the density, and the LER rises to 1.18–1.24 for hand-
683 written rules. Second, the best simple rule follows crop phenology. It shares
684 light across the middle of the season, around flowering and grain filling, and
685 produces electricity at the beginning and the end. It needs only the crop’s
686 thermal time, which is observable on the farm, and it is insensitive to forecast
687 errors, whereas the rule that reacts to heat and drought stress is worse in
688 every climate and degrades with forecast error. Third, the qualitative result
689 does not depend on the weakest part of the model. The simulated shade

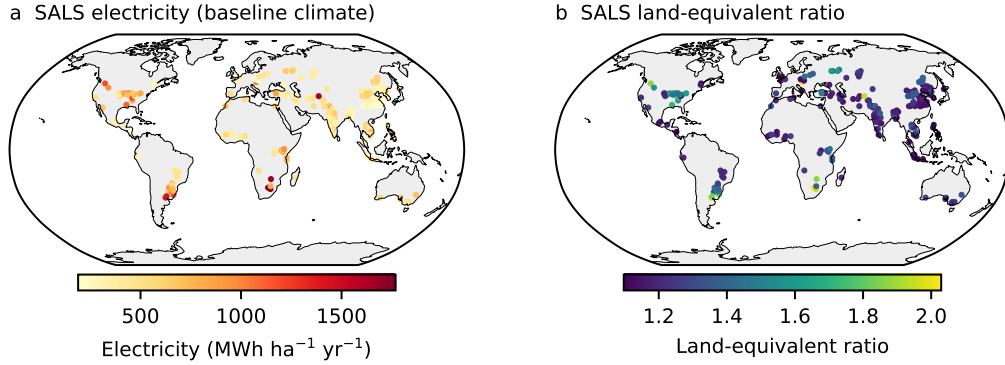


Figure 8: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

response is conservative against the field trials we could use (Section 3.6); calibrating it raises the allowed density, and the phenology rule stays the best method, with a gain over static arrays of 0.17, 0.19 and 0.30 LER under the conservative, calibrated and field-fitted responses. However, the response learned from the meta-analysis shows steeper losses for maize and grain legumes than the simulator reproduces (Section 3.7), so the absolute gains are lower bounds for cereals and potato and probably overestimates for maize and soybean. Fourth, a controller learned through the differentiable simulator is better than the best rule we could write, by 0.03–0.05 in LER and about 10% in electricity at equal retention, reproducibly over training seeds (range 0.002 in LER), three spatial splits, three canopy-cooling assumptions and four climates, and at 84–86% of the held-out sites. This gain is small compared with the gap between static and dynamic control.

4.2. What role does artificial intelligence play?

We tested three ways in which learning could matter, and the answers differ. The first is learning the crop’s response to shade from data. A hierarchical model fitted to 428 observations from 58 studies predicts a held-out study with an error 28% below that of proportional loss, and it shows that crop type is the information that transfers, whereas irradiation, climate zone and shade type add nothing measurable and a non-parametric model (gradient-boosted trees) is worse with this number of studies (Section 3.7). This is the part where learning clearly matters, because it replaces an assumption that

the main analysis shares with much of the literature by a response with quantified uncertainty, and it exposes a gap: the process-based simulator cannot reproduce the steep losses of maize and legumes. Learning the response was limited by the data, not by the model: two studies support the tuber curve, no rice study exists, and 46 panel observations are all leafy vegetables.

The second is calibration through the differentiable simulator. Gradient descent through whole seasons fitted the response of each crop to the learned curves and to the field trials, and it produced an ensemble of plausible responses that reconciles them (Section 3.8). The ensemble drives the design decision: a density chosen under the mean response meets the floor at 62–63% of sites if another plausible response holds, against 73–79% for an oracle, and a density chosen to meet the floor in 75% of plausible responses recovers most of that gap at a cost of 0.04 LER. The bottleneck of tracker-based agrivoltaics under a food-security floor is therefore the quantified uncertainty of the crop response that enters the design, and it is data on that response, not larger networks, that would improve the result. The guarantee holds only within the range of responses that the ensemble spans.

The third is control, and here learning contributes little. The learned controller adds 0.01–0.03 LER over the best hand-written rule, which follows crop phenology, under every response we tested, and a reader who needs a simple, auditable controller can use that rule and obtain most of the electricity gain over static arrays. Neither of the two extensions that we expected to matter under uncertainty did: training on an ensemble of responses did not change the policy, and neither did giving the controller a farm-measurable signal, the biomass of the crop relative to a reference plot, nor a feedback rule built on that signal. The learned policy also has to be initialised from the phenology rule (Section 4.5). We therefore do not claim that learned control is what makes agrivoltaics work: in this model, the schedule that a learned policy converges to is a smooth function of crop phenology, which a rule can already express.

4.3. What the results do not establish

The early hypothesis that stress-aware control, that is, shading the crop on heat and drought days, would be a main source of benefit under warming was not supported. The stress rule performs worse than the phenology rule in every climate, and a learned controller without stress inputs performs as well as one with them. A plausible reason is that in our crop model the yield gain from shading on a hot or dry day is smaller than the loss of light on

that day, so stress information does not help. Whether this holds for crop models with more detailed canopy-temperature and heat-stress responses is an open question. Agrivoltaics does not protect staple production against warming in our simulations: light-sharing systems produce about 90% of the open-field yield in every climate and follow the same decline in maize and rice (Fig. 6).

4.4. Food security in practice

Holding the historical mean retention at 90% does not hold it in each season or at each site. 70% of SALS seasons meet the floor, the lower tail is slightly heavier than for the rules, and production-weighted retention is below 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-constrained design raises season-level compliance to about 92% at a cost of 0.065 in LER, and the ranking of the controllers is preserved. Whichever formulation is chosen, a per-site regulatory floor would need a region- and crop-specific design step. The upscaled potential (Section 3.10) shows the trade-off in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply need about 10 TWp of modules, several times the present global fleet, and the best sites are rainfed maize and soybean areas where production-weighted retention is lowest.

4.5. Limitations

- *Simulation only.* All results come from a simulator. PV output was benchmarked against PVGIS (29 sites, mean bias -4.9%), but the crop response to shade was learned from one meta-analysis data set and compared with four field trials (Sections 3.6 and 3.7). The simulated response is conservative with respect to the trials (rice, wheat, potato, maize) but too mild for maize and soybean relative to the meta-analysis, where the 40%-shading relative yields are 0.42 and 0.40 against 0.75 and 0.48 in the simulator. The harvest-index sensitivities h are assumptions, and the sensitivity analysis uses rule controllers on a subsample of 22 sites.
- *Canopy cooling.* The effect of shade on canopy temperature was represented by one coefficient κ , applied to a heat-stress function that was calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018). The ranking of the controllers is robust to κ between 0 and 2°C, but the absolute electricity changes by about $\pm 8\%$. Nocturnal warming under modules and rain redistribution are not modelled.

- 785 • *Crop model.* SIMPLE has a fixed harvest index and linear radiation-
786 use efficiency. More detailed models with grain-number formation and
787 canopy energy balance may favour different schedules.
- 788 • *Climate scenarios.* Warming was applied as uniform temperature shifts
789 with fixed precipitation, radiation and relative humidity, and with un-
790 changed cultivars and sowing dates. It is not a bias-adjusted climate
791 projection.
- 792 • *Initialisation and state information.* The network is initialised from the
793 phenology rule, because training from random weights collapsed to full
794 sharing. The gain of SALS is therefore the gain of fine-tuning that rule,
795 and we cannot say whether a better optimiser or a richer policy class
796 would find larger gains. The controller uses the simulated state of the
797 crop and soil (thermal time, canopy interception, soil water) without
798 measurement error, and the forecast errors we tested apply to weather
799 only.
- 800 • *Economics and deployment.* There is no cost, market or grid analysis.
801 The value of midday electricity, curtailment, storage, structural cost
802 and farmer acceptance (Trommsdorff et al., 2021) will determine the
803 realisable share. The average row pitch at the chosen densities (about
804 13 m for 2.2-m modules) leaves room for machinery, but the densest
805 designs (GCR up to 0.5) have a pitch of only 4.4 m.

806 5. Conclusions

807 Light sharing, the daily choice between maximum electricity and letting
808 light through to the crop, is the main lever of tracker-based agrivoltaics un-
809 der a food-security floor. In a differentiable simulator of trackers, PV output,
810 crop light, microclimate and crop growth, evaluated fairly on 400 crop-site
811 pairs with the array density of each site chosen from its own history, light-
812 sharing rules allowed about three times the module density of static arrays
813 at a mean yield retention of 90%. They raised the land-equivalent ratio
814 from 1.07 to 1.18–1.24, and a rule that follows crop phenology and needs no
815 weather forecast was the best of them. The simulated shade response is con-
816 servative against the field trials we could use, so these gains are lower bounds,
817 and calibrating the response raised them without changing the ranking of
818 methods. The crop’s response to shade, the least certain input, was learned

819 from 428 field observations: a hierarchical model predicted held-out stud-
820 ies with an error 28% below that of proportional loss, and gradient descent
821 through the simulator turned it into an ensemble of plausible responses. The
822 simulator is too mild for maize and soybean, so the gains for those crops are
823 probably overestimated. A learned controller added a further 0.01–0.05 LER
824 by following a smooth phenology-dependent schedule that a rule can express;
825 training it on an ensemble of shade responses or giving it a farm-measurable
826 biomass signal added nothing. Where learning changed a decision was the
827 design: choosing the array density so that the floor holds in 75% of the plu-
828 sible responses raised the share of compliant sites from 62–63% to 71–73%
829 (oracle 73–79%) at a cost of 0.04 LER. The yield floor is met on average but
830 not in every season or region, and the electricity potential of 5% of staple
831 cropland (about 14 PWhyr⁻¹) requires about 10 TWp of modules. Future
832 work should validate the crop response to shade under trackers with more
833 field data, notably for maize and legumes where the meta-analysis and the
834 trials disagree, extend the crop model to grain-number formation and canopy
835 energy balance, add costs and grid constraints, and pursue controllers that
836 guarantee the food-security floor per site and per season.

837 **CRediT authorship contribution statement**

838 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal
839 analysis, Investigation, Data curation, Visualization, Writing – original draft,
840 Writing – review & editing.

841 **Declaration of competing interest**

842 The author has no known competing financial interests or personal re-
843 lationships that could have appeared to influence the work reported in this
844 paper.

845 **Data and code availability**

846 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
847 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA
848 POWER daily weather (Sparks, 2018) and the SIMPLE crop parameters
849 (Zhao et al., 2019). The code of the simulator, of SALS and of all experi-
850 ments, together with the site list, is available from the author’s public reposi-
851 tory (<https://github.com/vinhqdang/Artificial-Intelligence-in-Agriculture>).

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855 **Appendix A. Statistical comparison**

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Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS (ours)	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS (ours)	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS (ours)	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS (ours)	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: mean of reported seasons; for wheat and potato the two-year range is -18.7 to $+2.7\%$ and -18.2 to $+11\%$.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 5: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table 6: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with $s > 0$). Covariates: irradiation, climate zone, shade type, intercropping.

Model	RMSE	MAE	Median AE	Mean MAE per study
Proportional loss, $y = 1 - s$	0.430	0.305	0.210	0.224
One curve for all crops	0.372	0.265	0.193	0.208
Curve per crop type, no pooling	0.313	0.208	0.141	0.165
Curve per crop type, partial pooling	0.312	0.210	0.150	0.162
Partial pooling + covariates	0.311	0.206	0.148	0.161
Gradient-boosted trees	0.321	0.219	0.162	0.184

Table 7: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, mean covariates) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

Crop	Shading	Data-driven	Simulator, E1	Simulator, E2	Proportional
Wheat	20%	0.93 [0.88, 0.96]	0.88	0.92	0.80
Wheat	40%	0.72 [0.59, 0.78]	0.70	0.77	0.60
Rice	20%	0.93 [0.88, 0.96]	0.85	0.90	0.80
Rice	40%	0.72 [0.59, 0.78]	0.66	0.73	0.60
Maize	20%	0.73 [0.58, 0.86]	0.86	0.90	0.80
Maize	40%	0.42 [0.27, 0.59]	0.69	0.75	0.60
Soybean	20%	0.71 [0.65, 0.79]	0.77	0.74	0.80
Soybean	40%	0.40 [0.34, 0.53]	0.53	0.48	0.60
Potato	20%	0.93 [0.92, 0.97]	0.88	0.92	0.80
Potato	40%	0.71 [0.67, 0.83]	0.73	0.79	0.60

Table 8: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier h on the flowering-window harvest-index penalty (1 = assumed value).

Crop	c	h multiplier
Wheat	0.91 ± 0.31	0.74 ± 0.53
Rice	0.61 ± 0.25	0.83 ± 0.33
Maize	0.14 ± 0.11	0.16 ± 0.13
Soybean	0.05 ± 0.09	1.66 ± 0.38
Potato	0.62 ± 0.20	0.90 ± 0.50

Table 9: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out site, 21 sites). The proportional (conservative) and harsh responses lie outside the range of the calibrated ensemble.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule	1.164	1.190	1.456	1.122
SALS (trained on proportional response)	1.178	1.209	1.489	1.132
SALS-robust (ensemble-trained)	1.179	1.204	1.491	1.127
SALS-adaptive (ensemble, biomass feedback)	1.176	1.202	1.495	1.124

Table 10: Array design under shade-response uncertainty (ensemble E2). Truth: each of five ensemble members in turn. Designs: oracle (density chosen under the truth), ensemble mean (density chosen under the mean response) and uncertainty-aware (density chosen so that the floor is met in at least 75% of the *other* four members). Means over the five held-out members; retention is the mean season retention, site compliance the share of sites whose mean retention is ≥ 0.9 .

Design	Method	GCR	Retention	Seasons ≥ 0.9 (%)	Sites ≥ 0.9 (%)	LER
oracle	Phenology rule	0.261	0.913	76	79	1.412
oracle	SALS	0.256	0.912	73	73	1.438
ensemble mean	Phenology rule	0.272	0.907	60	63	1.427
ensemble mean	SALS	0.262	0.909	62	62	1.446
uncertainty-aware	Phenology rule	0.245	0.918	73	73	1.385
uncertainty-aware	SALS	0.238	0.917	69	71	1.406

Table A.11: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92